Cambridge Assessment International Education

# Cambridge IGCSE™

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*7179838283*

**CHEMISTRY** **0620/42**

Paper 4 Theory (Extended) **October/November 2020**

**1 hour 15 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].
- The Periodic Table is printed in the question paper.

This document has **12** pages. Blank pages are indicated.

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**[Turn over**

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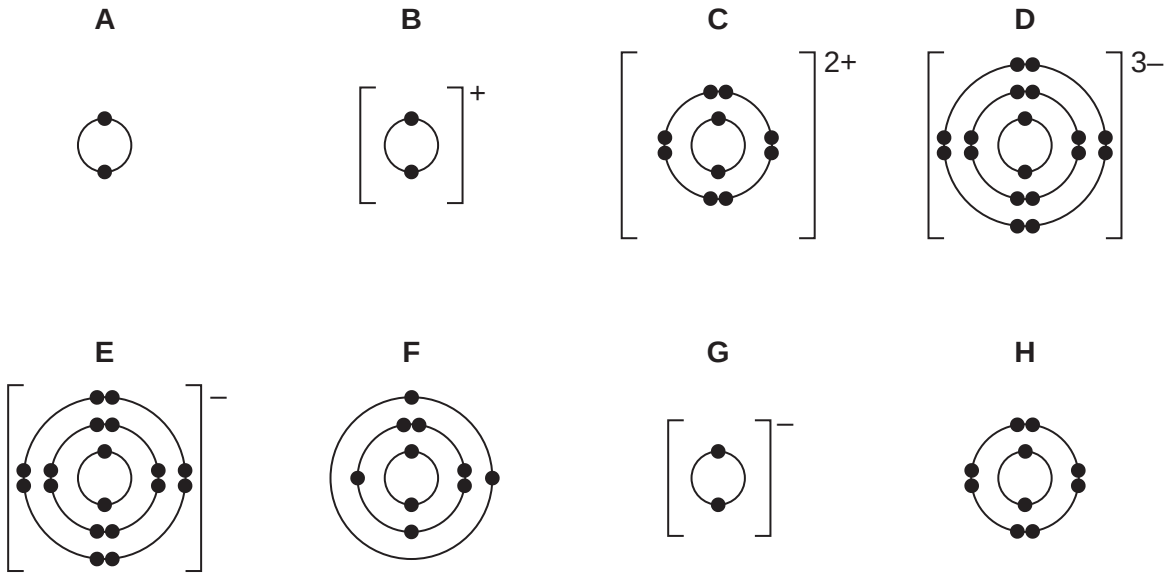
**1** The electronic structures of some atoms and ions are shown.




**(a)** Write the letters, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, of the electronic structures which show:

**(i)** atoms of two different noble gases ..... and ..... [2]

**(ii)** an ion of a Group I element ..... [1]

**(iii)** an ion of a Group V element ..... [1]

**(iv)** a pair of ions that could form a compound with the formula $XY_2$. ..... and ..... [1]

**(b)** State which electronic structure, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, is incorrect.

Explain why.

incorrect electronic structure .....

explanation .....

..... [2]

**(c)** State how many protons are found in the nucleus of ion **C**. ..... [1]

**(d)** Use the Periodic Table to deduce:

**(i)** the chemical symbol for ion **G** ..... [1]

**(ii)** the element which forms an ion with a 3+ charge and the same electronic structure as **H**.

..... [1]

[Total: 10]

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**2** Soluble salts can be made by adding a metal carbonate to a dilute acid.

**(a)** Give the formula of the dilute acid which reacts with a metal carbonate to form a nitrate salt.

..... [1]

**(b)** A student wanted to make hydrated iron(II) sulfate crystals, $FeSO_4 \bullet xH_2O$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

**step 1** Add dilute sulfuric acid to a beaker.

**step 2** Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

**step 3** Filter the mixture formed in **step 2**.

**step 4** Heat the filtrate until it is a saturated solution. Allow to cool.

**step 5** Once cold, pour away the remaining solution. Dry the crystals between filter papers.

**(i)** Why must the iron(II) carbonate be added in excess in **step 2**?

..... [1]

**(ii)** State **two** observations in **step 2** that would show that iron(II) carbonate was in excess.

1 .....

2 .....
[2]

**(iii)** Describe what should be done during **step 3** to ensure there is a maximum yield of crystals.

..... [1]

**(iv)** A saturated solution is formed in **step 4**.

Describe what a saturated solution is.

.....

..... [2]

**(v)** Name a different compound that could be used instead of iron(II) carbonate to produce hydrated iron(II) sulfate crystals from dilute sulfuric acid.

..... [1]

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**(c)** On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $FeSO_4 \bullet xH2O$, had a mass of 278 g.

Determine the value of *x* using the following steps:

- calculate the mass of one mole of $FeSO_4$

mass = ..... g

- calculate the mass of $H_2O$ present in one mole of $FeSO_4 \bullet xH_2O$

mass of $H_2O$ = ..... g

- determine the value of *x*.

*x* = .....
[3]

**(d)** Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

**step 1** Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

**step 2** Filter the mixture formed in **step 1**.

**step 3** Dry the residue.

**(i)** State the term used to describe this method of making salts.

..... [1]

**(ii)** Give the observation the student would make during **step 1**.

..... [1]

**(iii)** Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

..... [3]

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**(e)** Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas.

The equation for this reaction is shown.

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

Calculate the volume of chlorine gas, in $cm^3$, that reacts to form 2.34 g of NaC*l*.

The reaction takes place at room temperature and pressure.

volume of chlorine gas = ..... $cm^3$ [3]

**(f)** Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

**(i)** Explain why, in terms of structure and bonding.

.....

.....

.....

..... [3]

**(ii)** Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

..... [1]

**(iii)** State the type of change that occurs at the positive electrode in **(ii)**.

Explain your answer in terms of electron transfer.

type of change .....

explanation .....
[2]

**(iv)** Describe what else can be done to sodium chloride to allow it to conduct electricity.

..... [1]

[Total: 26]

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**3** Group I metals are very reactive. Transition elements are also metals but are less reactive than Group I metals.

**(a)** State **two** physical properties of Group I metals which are similar to those of transition metals.

1 .....

2 .....
[2]

**(b)** Describe **two** ways in which the physical properties of Group I metals are different from those of transition metals.

1 .....

.....

2 .....

.....
[2]

**(c)** When Group I metals are added to water they fizz and an alkaline solution forms.

**(i)** Name the gas given off.

..... [1]

**(ii)** Identify the ion present in the solution which makes the solution alkaline.

..... [1]

**(iii)** Write the chemical equation for the reaction between sodium and water.

..... [2]

**(d)** When the transition element iron is added to water the iron rusts.

When an iron object is coated with a layer of zinc, rusting is prevented.

**(i)** Name this process of coating iron objects with a layer of zinc.

..... [1]

**(ii)** Explain how completely coating an iron object with a layer of zinc prevents rusting.

..... [1]

**(iii)** Rusting of iron ships can be prevented by attaching zinc blocks to the hull of the ship.

Explain how this prevents rusting.

.....

..... [2]

[Total: 12]

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**4** Alkenes and alkanes are homologous series of compounds containing carbon and hydrogen atoms.

**(a)** State the name of the type of compound made from carbon and hydrogen atoms only.

..... [1]

**(b)** Alkenes take part in addition reactions.

**(i)** Describe what is meant by the term *addition reaction*.

..... [1]

**(ii)** Draw the structure of the product made in the addition reaction between propene and bromine. Show all of the atoms and all of the bonds.

[2]

**(iii)** Describe the colour change seen when propene is added to aqueous bromine.

from ..... to ..... [2]

**(iv)** Draw the structures of molecules of **two** different alkenes which both undergo an addition reaction with steam to form butan-2-ol. Show all of the atoms and all of the bonds.

[2]

**(c)** Propane undergoes a substitution reaction with chlorine.

Write the chemical equation for the reaction between one molecule of propane and one molecule of chlorine.

..... [2]

[Total: 10]

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**5** This question is about alcohols, carboxylic acids and esters.

**(a)** Ethanol will react with hot aqueous potassium manganate(VII) to form ethanoic acid.

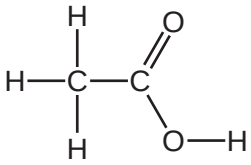
**(i)** State the other condition needed for this reaction to take place.

..... [1]

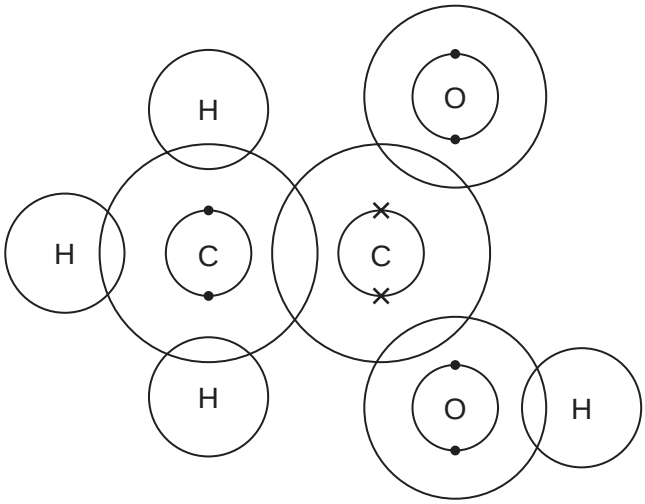
**(ii)** State the type of chemical change that happens to the ethanol during this reaction.

..... [1]

**(iii)** The structure of ethanoic acid is shown.




Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanoic acid.




[3]

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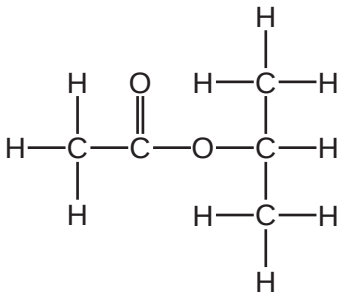
**(b)** Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Complete the table to show the similarities and differences in the properties of samples of these two acids of equal concentration.

| | dilute ethanoic acid | dilute hydrochloric acid |
|---|---|---|
| extent of dissociation | | |
| colour after adding universal indicator solution | | |
| observation when magnesium ribbon is added | | |

[6]

**(c)** Ethanoic acid will react with an alcohol to form the ester shown.



**(i)** Name the **other** product formed when ethanoic acid reacts with an alcohol to make this ester.

..... [1]

**(ii)** Give **one** condition needed when ethanoic acid reacts with the alcohol to make this ester.

..... [1]

**(iii)** Draw the structure of the alcohol which was added to ethanoic acid to make this ester. Show all of the atoms and all of the bonds.

[2]

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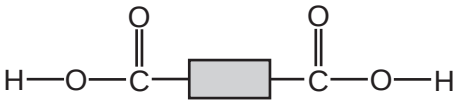
**(d)** Polyesters can be manufactured from carboxylic acids and alcohols.

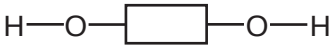
Hexanedioic acid has the structure: $HOOC–CH_2–CH_2–CH_2–CH_2–COOH$.

This structure can be simplified as shown.



Ethanediol has the structure: $HO–CH_2–CH_2–OH$.

This structure can be simplified as shown.



The functional groups are found at the end of each molecule.

**(i)** State what is meant by the term *functional group*.

..... [1]

**(ii)** Determine the empirical formula of hexanedioic acid.

..... [1]

**(iii)** Calculate the percentage by mass of oxygen present in ethanediol.

Give your answer to the nearest whole number.

..... % [2]

**(iv)** Complete the diagram to show a section of polyester manufactured from hexanedioic acid and ethanediol. Include all of the atoms and all of the bonds in the linkages.

[2]

**(v)** State the name of a polyester.

..... [1]

[Total: 22]

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## The Periodic Table of Elements

**Key**

| atomic number |
|---|
| atomic symbol |
| name |
| relative atomic mass |

Group

| I | II | | | | | | | | | | | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1<br>H<br>hydrogen<br>1 | | | | | | | | | | 2<br>He<br>helium<br>4 |
| 3<br>Li<br>lithium<br>7 | 4<br>Be<br>beryllium<br>9 | | | | | | | | | | | 5<br>B<br>boron<br>11 | 6<br>C<br>carbon<br>12 | 7<br>N<br>nitrogen<br>14 | 8<br>O<br>oxygen<br>16 | 9<br>F<br>fluorine<br>19 | 10<br>Ne<br>neon<br>20 |
| 11<br>Na<br>sodium<br>23 | 12<br>Mg<br>magnesium<br>24 | | | | | | | | | | | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28 | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32 | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40 |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40 | 21<br>Sc<br>scandium<br>45 | 22<br>Ti<br>titanium<br>48 | 23<br>V<br>vanadium<br>51 | 24<br>Cr<br>chromium<br>52 | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56 | 27<br>Co<br>cobalt<br>59 | 28<br>Ni<br>nickel<br>59 | 29<br>Cu<br>copper<br>64 | 30<br>Zn<br>zinc<br>65 | 31<br>Ga<br>gallium<br>70 | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75 | 34<br>Se<br>selenium<br>79 | 35<br>Br<br>bromine<br>80 | 36<br>Kr<br>krypton<br>84 |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89 | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93 | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>– | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108 | 48<br>Cd<br>cadmium<br>112 | 49<br>In<br>indium<br>115 | 50<br>Sn<br>tin<br>119 | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127 | 54<br>Xe<br>xenon<br>131 |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137 | 57–71<br>lanthanoids | 72<br>Hf<br>hafnium<br>178 | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184 | 75<br>Re<br>rhenium<br>186 | 76<br>Os<br>osmium<br>190 | 77<br>Ir<br>iridium<br>192 | 78<br>Pt<br>platinum<br>195 | 79<br>Au<br>gold<br>197 | 80<br>Hg<br>mercury<br>201 | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207 | 83<br>Bi<br>bismuth<br>209 | 84<br>Po<br>polonium<br>– | 85<br>At<br>astatine<br>– | 86<br>Rn<br>radon<br>– |
| 87<br>Fr<br>francium<br>– | 88<br>Ra<br>radium<br>– | 89–103<br>actinoids | 104<br>Rf<br>rutherfordium<br>– | 105<br>Db<br>dubnium<br>– | 106<br>Sg<br>seaborgium<br>– | 107<br>Bh<br>bohrium<br>– | 108<br>Hs<br>hassium<br>– | 109<br>Mt<br>meitnerium<br>– | 110<br>Ds<br>darmstadtium<br>– | 111<br>Rg<br>roentgenium<br>– | 112<br>Cn<br>copernicium<br>– | | 114<br>Fl<br>flerovium<br>– | | 116<br>Lv<br>livermorium<br>– | | |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140 | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>– | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165 | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169 | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175 |
| actinoids | 89<br>Ac<br>actinium<br>– | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238 | 93<br>Np<br>neptunium<br>– | 94<br>Pu<br>plutonium<br>– | 95<br>Am<br>americium<br>– | 96<br>Cm<br>curium<br>– | 97<br>Bk<br>berkelium<br>– | 98<br>Cf<br>californium<br>– | 99<br>Es<br>einsteinium<br>– | 100<br>Fm<br>fermium<br>– | 101<br>Md<br>mendelevium<br>– | 102<br>No<br>nobelium<br>– | 103<br>Lr<br>lawrencium<br>– |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).

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